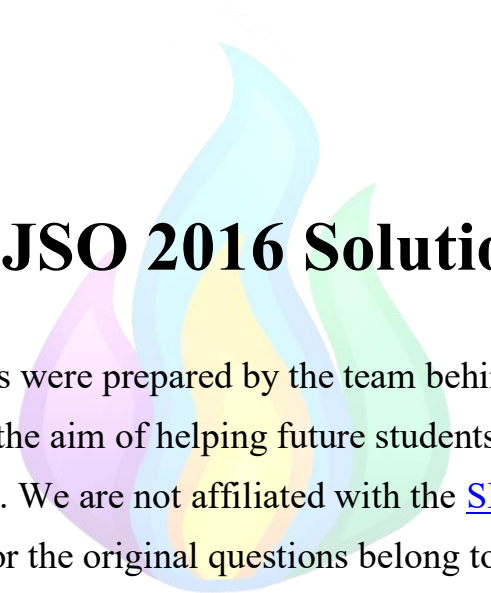




SLJSO 2016 Solutions

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Question No.	Field	References
<u>1</u>	Botany	<u>Transport in Plants</u>
<u>2</u>	Cell Biology	<u>Cell Structure and Organelles</u>
<u>3</u>	Human Biology	<u>Blood and its Components</u>
<u>4</u>	Ecology	<u>Ecological Pyramids</u>
<u>5</u>	Human Biology	<u>Endocrine System</u>
<u>6</u>	Evolutionary Development	<u>Evolution</u>
<u>7</u>	Human Biology	<u>Blood and its Components</u>
<u>8</u>	Human Biology	<u>Sensory Organs</u>
<u>9</u>	Genetics	<u>Genetic Disorders</u>
<u>10</u>		
<u>11</u>	Botany	<u>Phytohormones and Tropism</u>
<u>12</u>	Human Biology	<u>Nervous System</u>
<u>13</u>	Human Biology	<u>Muscular System</u>
<u>14</u>	Human Biology	<u>Blood and its Components</u>
<u>15</u>		
<u>16</u>	Chemical Structures	<u>Electron Configurations</u>
<u>17</u>	Chemical Structures	<u>The Periodic Table</u>
<u>18</u>	Chemical Structures	<u>The Periodic Table</u>
<u>19</u>	Chemical Structures	<u>Trends in the Periodic Table</u>
<u>20</u>	Chemical Structures	<u>Types of Bonding</u>
<u>21</u>	Chemical Structures	<u>Polarity of Bonds</u>
<u>22</u>		
<u>23</u>	Electrochemistry	<u>Activity Series of Metals</u>
<u>24</u>	Chemical Equilibrium	<u>Solubility Equilibria</u>
<u>25</u>	Electrochemistry	<u>Redox Reactions</u>
<u>26</u>	Electrochemistry	<u>Redox Reactions</u>
<u>27</u>	Thermodynamics	<u>Ideal Gas Law</u>

<u>28</u>	Laboratory Guide	Precipitates
<u>29</u>	Chemical Kinetics	<u>Activation Energy</u>
<u>30</u>	Electrochemistry	<u>Faraday's Law</u>
<u>31</u>	Mechanics	<u>Torque</u>
<u>32</u>	Electromagnetism	<u>Ohm's Law</u> <u>Series and Parallel Circuits</u>
<u>33</u>	Mechanics	<u>Kinematics Formulas</u>
<u>34</u>	Mechanics	<u>Newton's Laws</u>
<u>35</u>	Thermodynamics	<u>Ideal Gas Law</u>
<u>36</u>	Electromagnetism	<u>Series and Parallel Circuits</u>
<u>37</u>	Mechanics	<u>Motion Graphs</u>
<u>38</u>	Light	<u>Reflection</u>
<u>39</u>	Mechanics	<u>Pressure</u>
<u>40</u>	Mechanics	<u>Buoyancy</u>
<u>41</u>	Mechanics	<u>Newton's Laws</u>
<u>42</u>	Algebra	
<u>43</u>	Areas	
<u>44</u>	Thermodynamics	<u>Ideal Gas Law</u>
<u>45</u>	Areas	
<u>46</u>	Algebra	
<u>47</u>	Mechanics	<u>Buoyancy</u>
<u>48</u>	Light	<u>Refraction</u>
<u>49</u>	Mechanics	<u>Pressure</u>
<u>50</u>	Thermodynamics	<u>Ideal Gas Law</u>

Question 1

The main function of the xylem is the transportation of dissolved minerals and water within plants; the phloem is responsible for the food transportation. The thickness growth in plants is issued by the cambium, not the xylem.

Correct Answer: C

Question 2

The Golgi bodies are mainly processing and packaging proteins from the ER (endoplasmic reticulum) for further transportation. Lysosomes break down waste and foreign substances within cells. Ribosomes are enzymes that synthesize proteins by translating mRNA (messenger RNA) into amino acid chains. The nucleus stores the DNA and thus carries the genetic information to transmit hereditary characteristics from one generation to the next.

Correct Answer: B

Question 3

Red blood cells are responsible for carrying oxygen. Lymphocytes are part of our immune system and help us fight infections and diseases. Hemoglobin transports oxygen from the lungs and carbon dioxide from the tissues back to the lungs. The main function of platelets is helping with blood clotting.

Correct Answer: C

Question 4

The third trophic level of an environmental number pyramid is secondary consumers, which eat primary consumers and thus are carnivores or omnivores.

Correct Answer: D

Question 5

The thyroid produces hormones such as thyroxine (T4) and triiodothyronine (T3). The pituitary makes hormones that regulate body functions, e.g., growth hormones (GH) or follicle-stimulating hormones (FSH). The adrenal gland mainly produces cortisol and adrenaline. In this task, the only gland producing digestive enzymes is the pancreas; the hormones are, e.g., amylase, trypsin, and lipase.

Correct Answer: C

Question 6

Similarities within a group of animals are mostly derived from a common ancestor.

Correct Answer: C

Question 7

Red blood cells do not contain a nucleus. Because of that, blood transportation is optimized due to a higher percentage of hemoglobin within the cells. The remaining blood cells contain a nucleus.

Correct Answer: C

Question 8

The cochlea's role is converting sound waves to electric signals that our brain can interpret. The tympanic membrane separates our outer ear from the middle ear. The eustachian tube is a passageway that connects the middle ear to the throat and nose. Only the semicircular canals help maintain equilibrium by sensing rotational and angular movements of the head.

Correct Answer: B

Question 9

A goiter is a lump or swelling at the front of the neck caused by a swollen thyroid, which can affect both genders. Malaria is a disease caused by a parasite. A chronic kidney disease of unknown etiology can be caused by genetic factors, but not necessarily. Color blindness is a hereditary disease affecting mainly men because it's coded on the X chromosome.

Correct Answer: C

Question 10

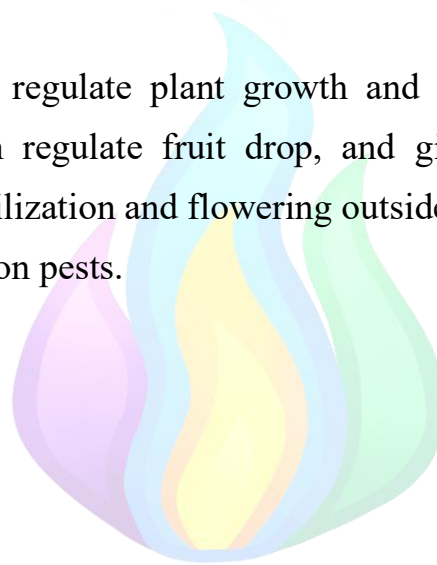
Having sunken stomata indicates that a plant is located at a hot and dry site. Sunken stomata decrease water loss due to less transpiration.

Correct Answer: C

Question 11

Plant growth substances regulate plant growth and development. For example, ethylene and auxins can regulate fruit drop, and gibberellins can trigger fruit development without fertilization and flowering outside the season. Plant hormones do not have toxic effects on pests.

Correct Answer: D



Question 12

Motor nerves carry impulses from the central nervous system to muscles and glands. Sensory neurons carry impulses from sensory organs to the central nervous system. Interneurons link sensory and motor neurons.

Correct Answer: C

Question 13

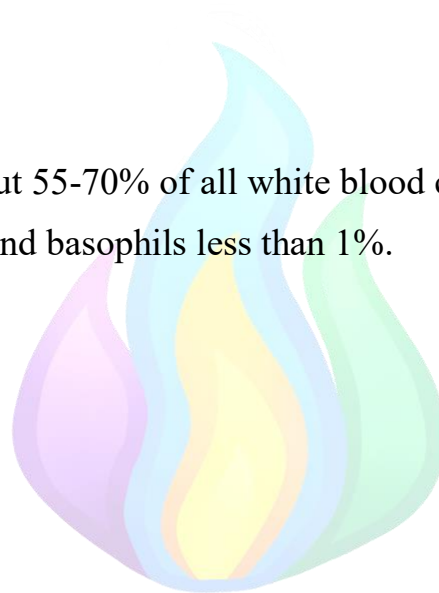
Heart muscles are involuntary, striated muscles. Skeletal muscles are striated and voluntary; smooth muscles are involuntary and unstriated.

Correct Answer: B

Question 14

Neutrophils make up about 55-70% of all white blood cells, lymphocytes about 20-40%, eosinophils 1-6%, and basophils less than 1%.

Correct Answer: C



Question 15

Endemic animals are animals that are found only in one geographic area. The red-faced malkoha is found only in Sri Lanka. Elephants and spotted deer are obviously not endemic to Sri Lanka; the Asian koel lives in wide parts of Asia and Australia.

Correct Answer: B

Question 16

The same electron structure as the chloride ion means having the same number of electrons as it. Because the ion is a monovalent anion, the number of electrons in it is $N = Z(\text{Cl}) + 1 = 18$.

A - The Ca^{2+} ion has $Z - 2 = 20 - 2 = 18$ electrons.

B - The chlorine atom has $Z = 17$ electrons.

C - The Na^+ ion has $Z - 1 = 11 - 1 = 10$ electrons.

D - The potassium atom has $Z = 19$ electrons.

Correct Answer: A

Question 17

Elements in Group III are generally trivalent (having the electron structure ns^2np^1). Trivalent elements form oxides with the general formula X_2O_3

Correct Answer: D

Question 18

Elements in the same group are characterized by the same valence (and the same outer shell structure) but an increasing number of electron shells. So, the number of electrons in the outermost energy level (the valence shell) remains constant when moving down in a group.

Correct Answer: B

Question 19

The reactivity of sodium is given by the fact that it easily gets oxidized.



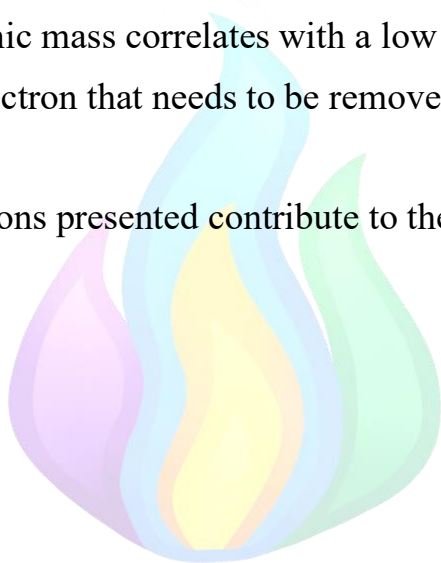
A - Sodium is very reactive because its ionization energy is low.

B - The reactivity of sodium also comes from the fact that, to reach a stable octet, sodium only must lose one electron (it has only one outermost electron).

C - Its relatively low atomic mass correlates with a low nuclear charge, so a weaker attraction is felt by the electron that needs to be removed to form Na^+ ions.

In conclusion, all the reasons presented contribute to the high reactivity of sodium.

Correct Answer: D



Question 20

Ionic bonds require an anion and a cation. Typically, cations are formed by metals, while anions are formed by nonmetals. The only compound in the given list that contains a metal is magnesium fluoride. The others are covalent compounds, created by covalent bonding between nonmetals.

Correct Answer: B

Question 21

- (A) HBr is a polar molecule that has a dipole moment.
- (B) CH₄ has slightly polar bonds but a symmetric tetrahedral structure, so the dipole moments (which were anyway very small) cancel out.
- (C) H₂ only has a homonuclear bond (a covalent bond between two identical atoms), which has no dipole moment (neither of the hydrogen atoms is more electronegative than the other).
- (D) CO₂ has polar C=O bonds but a linear structure, so the dipole moments of the bonds cancel out.

Correct Answer: A

Question 22

Milk is an emulsion (small fatty particles are dispersed in an aqueous medium). The fact that fat molecules are not found in between the water molecules means milk is not a homogenous mixture (even though it appears like this on a macroscopic scale) but rather a heterogeneous one.

Correct Answer: D

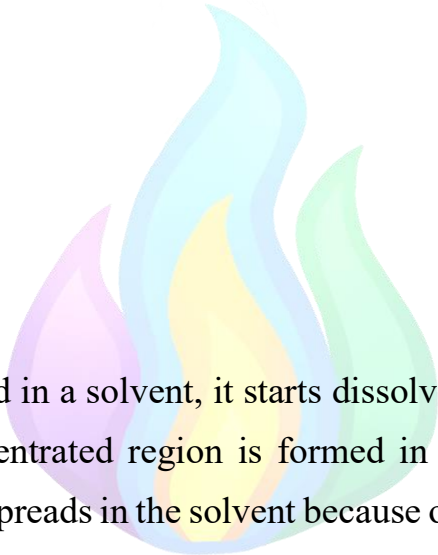
Question 23

For metal A to be able to reduce metal B (found in an aqueous solution in the form of a salt), we need to have the relationship $A > B$ between the reactivity of the two metals (also, A needs to have a smaller reducing potential than B).

- (A) Zn is more reactive than Cu, so it does displace it from its salt.
- (B) Fe is more reactive than Ag, so it does displace it from its salt.
- (C) Cu is not more reactive than Al, so it cannot displace it from its salt.
- (D) Mg is more reactive than Al, so it does displace it from its salt.

Correct Answer: C

Question 24

- 
- (A) When a solid is placed in a solvent, it starts dissolving. Because of this, around the solid particle, a concentrated region is formed in which the solid can hardly dissolve. The solute then spreads in the solvent because of Brownian motion. Stirring the solution makes this process go faster and helps the dissolution process.
 - (B) Pulverizing the solid means dividing it into more and smaller particles, which dissolve faster and more evenly spread across the volume of the solvent.
 - (C) Heating the solution increases Brownian motion and the possibility of diffusing the solute between solvent molecules, making the dissolution process easier.
 - (D) Increasing the pressure increases the solubility of gases, but in the case of solids, it does not affect solubility.

Correct Answer: C

Question 25

On the products side of the equation, there are 6 nitrogen atoms in copper nitrate and 2 nitrogen atoms in nitrogen monoxide, 8 N in total. To balance the number of nitrogen atoms, we add the coefficient 8 to nitric acid.

On the reactants side of the reaction, there are now 8 hydrogen atoms. All of these are found in water after the reaction. Because every water molecule has 2 hydrogen atoms, the coefficient of water is 4.

Correct Answer: C

Question 26

In hydrogen sulfide, the oxidation state of sulfur is -2. In Br_2 , the oxidation state of bromine is 0. In the products, the oxidation state of sulfur in sulfuric acid is +6, while that of bromine in hydrobromic acid is -1.

Thus, in the reactions, the oxidation state of sulfur increases and that of bromine decreases. The oxidizing agent gets reduced (and it oxidizes the other substance). Getting reduced means gaining electrons and having a decrease in the oxidation state. In this reaction, this is the case of bromine, so the oxidizing agent is bromine.

Correct Answer: B

Question 27

Pressure remains constant = isobaric transformation.

In the case of an isobaric transformation, we can use Charles's law: $\frac{V}{T} = \text{constant}$.

We can write $\frac{V_1}{T_1} = \frac{V_2}{T_2}$, so $V_2 = V_1 \frac{T_2}{T_1}$

Using $T_1 = 25 + 273 = 298\text{K}$ and $T_2 = 100 + 273 = 373\text{K}$ and $V_1 = 300\text{ mL}$, we get $V_2 = 376\text{ mL}$.

Correct Answer: C

Question 28

In general, all nitrates are soluble. In fact, the most used silver salt is silver nitrate. Silver halides (in the case of this question, bromide and chloride) are white precipitates, turning more yellowish when the halide is heavier (AgCl —white, AgI —yellowish white). Silver carbonate is a white precipitate.

Correct Answer: B

Question 29

In the given energy profile, the reactants are represented by A, the products by C, and the transition state by B.

The activation energy is the energy required to start the reaction—that is, the energy necessary to form the transition state (the state in which the energy is maximum).

From the energy profile given, the activation energy is the energy of the transition state minus that of the reactants, so B-A.

Correct Answer: D

Question 30

The current is constant, so the amount of copper being oxidized/reduced per unit of time is constant, so the graph is linear. All the negative ions gather at the copper anode (positive electrode). These are HO^- and SO_4^{2-} . The species that could be oxidized (oxidation takes place at the anode) are Cu or the two anions. Because of the higher oxidation potential, Cu is the one being oxidized. Copper is turned into ions and dissolved in the solutions, so the mass of the anode decreases.

We could have concluded that the anions are not being oxidized from the fact that that would have resulted in a constant mass of the anode, and this variant does not exist.

Correct Answer: D

Question 31

This is a problem in statics, so we should use the equality of the torques.

$$m_p \times g \times x = (m_q + m_r) \times g \times 3x$$

$$m_q \times g \times y = m_r \times g \times 2y$$

Which simplifies to

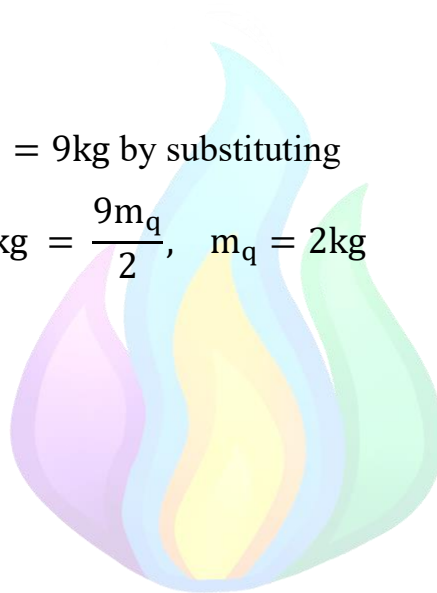
$$m_p = 3(m_q + m_r)$$

$$m_q = 2m_r$$

And with the info that $m_p = 9\text{kg}$ by substituting

$$m_p = 3(m_q + \frac{m_q}{2}), \quad 9\text{kg} = \frac{9m_q}{2}, \quad m_q = 2\text{kg}$$

Correct Answer: B



Question 32

As a common trick in circuit problems, we'll simplify by substituting parallel and series resistors.

Both the top and the bottom of the “square” add up to 5Ω , which totals to 2.5Ω for the square. Then adding the 1.5 resistor to the square gives us the total resistance of 4Ω . Using Ohm's law, we get $I = \frac{V}{R}$, $I = \frac{4V}{4\Omega} = 1\text{A}$.

Correct Answer: D

Question 33

It's clear that it is intended for us to get the acceleration, the distance, the force, and then the work.

$$a = \frac{v}{t} = 20 \frac{\text{m}}{\text{s}} \div 200\text{s} = 0.1 \frac{\text{m}}{\text{s}^2}$$

$$F = ma = 1000\text{kg} \times 0.1 \frac{\text{m}}{\text{s}^2} = 100\text{N}$$

$$s = \frac{vt}{2}, \quad s = 2000 \text{ m}$$

$$A = F \times s = 200 \text{ kJ}$$

Correct Answer: D

Question 34

We can just look at the mass that's in the air.

$$F_{\text{string}} - mg = 0$$

We are given in the picture that the weight (mg) is 4 N.

Therefore, the tension is also 4 N.

Correct Answer: A



Question 35

Using Boyle's law (or the ideal gas law)

$$P_1 V_1 = P_2 V_2$$

Initial air column length: $L_1 = 33\text{cm}$

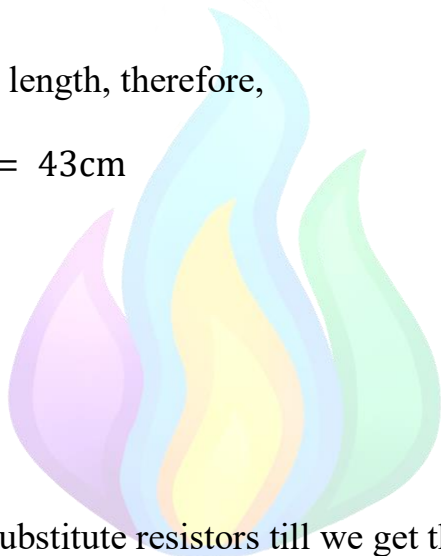
Initial pressure: 86 cm Hg (= atmospheric + 10 additional)

Final pressure after flipping: 66 cm Hg (= atmospheric - 10)

Volume is proportional to length, therefore,

$$L_2 = \frac{P_1}{P_2} L_1 = \frac{86}{66} \times 33 = 43\text{cm}$$

Correct Answer: D



Question 36

Same as in Question 32, substitute resistors till we get the equivalent resistor (R_{eq}).

The top line equates to $R + \frac{R}{2} + \frac{2R}{2} + \frac{R}{2} + R = 4R$.

And parallel with the bottom gives $\frac{1}{R_{eq}} = \frac{1}{4R} + \frac{1}{R} \rightarrow R_{eq} = 0.8 \Omega$

****Or we could've just looked at the picture, realized it's a parallel link with one branch being R , therefore the whole being $<1R$, saw the given answers, and realized that only one is under $1R$. This is a neat trick that I suggest you should use as much as possible; it saves valuable time. :)**

Correct Answer: D

Question 37

The simplest solution is just noticing that because there is no friction, there will be no slowing down; therefore, the only valid answer is the one that has no negatively sloped lines—C.

Correct Answer: C

Question 38

This is just simple geometry if we use that the angle of incidence is equal to the angle of refraction. ($\theta_r = \theta_i$)

There are many ways that we could do the angle chasing that give the formula:

$$\beta = 90^\circ - 26^\circ = 64^\circ$$

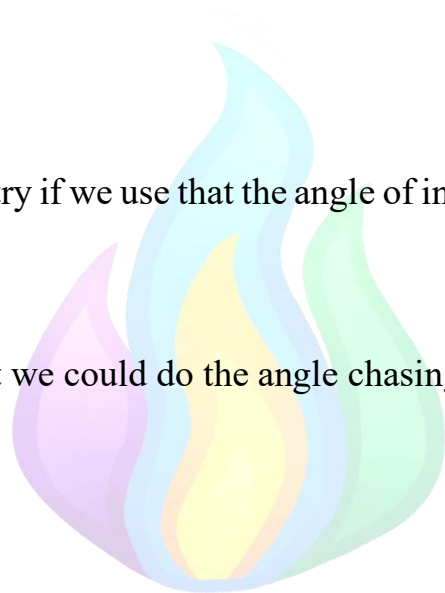
Correct Answer: D

Question 39

Using Pascal's law, the pressure on the liquid from the confined air equals the pressure on the same level on the right side of the U-tube. Then, using hydrostatic pressure and accounting for the atmosphere:

$$P = P_{\text{atm}} + \rho gh$$

Correct Answer: C



Question 40

We are given $mg = 15\text{N}$, $m = 1.5\text{kg}$ and $mg - F_U = 12\text{N}$

Using Archimedes' principle, we get $F_U = \rho Vg$

And by substituting, we get: $\rho Vg = 15\text{N} - 12\text{N}$, $V = 3\text{N}/(\rho g)$

$$V = 3 \times 10^{-4} \text{ m}^3, \quad \rho_o = \frac{m}{V} = 5000 \frac{\text{kg}}{\text{m}^3}$$

Correct Answer: B

Question 41

We could calculate the forces and add them and do a lot of work, but instead just look at the resulting forces in the y-axis: the gravitational force of 5 N downwards and the resulting force of the Ws upwards. The ball is stationary; therefore, $F_r = 5\text{N}$

Correct Answer: D

Question 42

The two equations are similar. So, we can write

$$y^2 = 4x = y^2 = (2x)^2 = 4x^2$$

$$4x = 4x^2$$

Hence, $x = 0$ or $x = 1$.

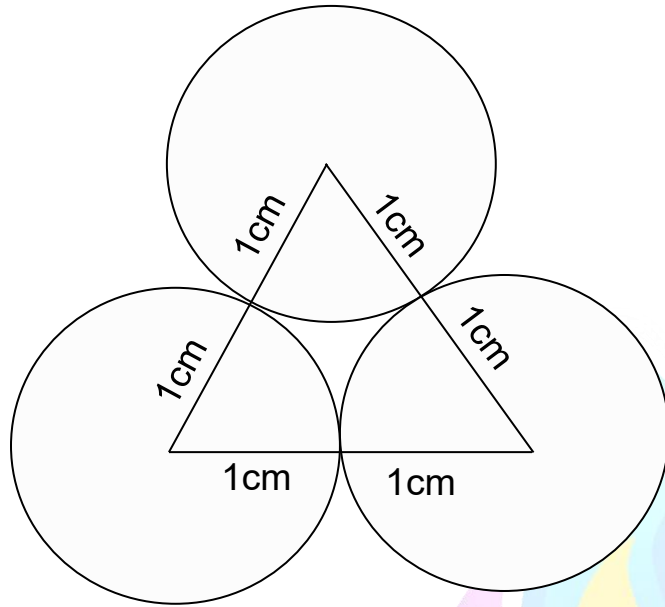
When $x=0$, $y = 0$ and when $x=1$, $y = 2$.

So, $(0,0)$ and $(1,2)$ are the only possible solutions.

Correct Answer: D

Question 43

We should do a little construction.



The corners of the triangles are the centers of the circles. After drawing this triangle, it is quite obvious that we could subtract the area of the 3 sectors from the area of the triangle.

$$\text{Area of the equilateral triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

$$\text{Base} = 2 \text{ cm. Height} = 2 \text{ cm} \times \sin 60 = 2 \text{ cm} \times 0.9 = 1.8 \text{ cm}$$

$$\text{Area} = \frac{1}{2} \times 2 \text{ cm} \times 1.8 \text{ cm} = 1.8 \text{ cm}^2$$

$$\text{Area of 1 sector} = \frac{60}{360} \times \pi r^2 = \frac{1}{6} \times 3 \times 1^2 = 0.500 \text{ cm}^2$$

$$\text{Area of 3 sectors} = 0.500 \text{ cm}^2 \times 3 = 1.500 \text{ cm}^2$$

$$\text{Area of the shaded portion} = 1.8 \text{ cm}^2 - 1.500 \text{ cm}^2 = 0.300 \text{ cm}^2$$

Correct Answer: D

Question 44

We are given: $V_2 = \frac{V_1}{3}$, $P_2 = 9P_1$, $T_1 = 300K$

Using the ideal gas law we get:

$$T_2 = T_1 \times \frac{P_2 V_2}{P_1 V_1} = 300K \times 3 = 900K = 627^\circ C$$

Correct Answer: C

Question 45

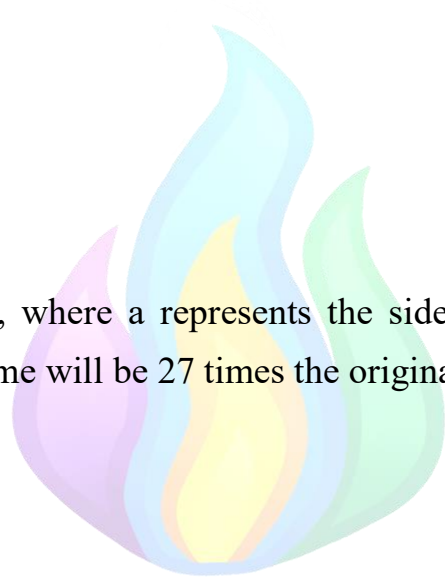
Volume of a cube = a^3 , where a represents the side length. So, when the side length is tripled, the volume will be 27 times the original volume.

Correct Answer: D

Question 46

If two lines are parallel, their gradient must be similar. The gradient of the graph $y=4x$ is 4. The gradient of the new line must also be equal to 4. The only option with an equation of a line where the gradient is 4 is option C.

Correct Answer: C



Question 47

Archimedes' principle: $F_b = \rho V g = m_{\text{liquid displaced}} g$

But since the block is floating, it is also equal to its weight...

2 CORRECT ANSWERS!?

Correct Answer: A, B

Question 48

Just like Newton's prism—refraction.

Correct Answer: D

Question 49

$$P - P_0 = \rho h g, \quad \rho = \frac{P - P_0}{h g} = 40 \text{ kPa} \div 20 \frac{\text{m}^2}{\text{s}^2} = 2000 \frac{\text{kg}}{\text{m}^3}$$

Correct Answer: D

Question 50

Reducing the amount of air reduces its pressure (simple logic or the ideal gas law); therefore, the outside greater pressure crushes the can.

Correct Answer: B